Aitken acceleration in Norma: two formulations

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Abstract

This note records the two Aitken-acceleration formulations implemented in Norma's non-overlapping Schwarz alternating solver, together with the displacement-form kinematic recovery they use. The purpose is to fix the exact formulas and index conventions so they can be reproduced in Sierra/SM. Both formulations are exposed through the multi-domain controller: the recursive Irons–Tuck variant (`relaxation: aitken`) and the non-recursive secant variant (`relaxation: aitken secant`). They are mathematically equivalent in exact arithmetic but differ in their initialization bookkeeping and in how they treat the interface velocity and acceleration.

1 Setting and fixed-point reformulation

Consider a domain decomposition of Ω into two non-overlapping subdomains Ω_1 and Ω_2 sharing the interface Γ. Norma drives the coupling *multiplicatively* (Gauss–Seidel ordering of the subdomain solves) with a strongly imposed Dirichlet condition on the Dirichlet side of the interface and a traction (Neumann) condition on the other. Let $\boldsymbol{g}$ denote the interface unknown, i.e. the displacement datum imposed on the interface. Following Sambataro and Tezaur [1] we write the multiplicative Schwarz sweep as a fixed-point map

$$\boldsymbol{g} \;\mapsto\; T(\boldsymbol{g}), \tag{1}$$

where $T(\boldsymbol{g})$ is obtained by solving the subdomains in order with the current interface data and projecting the resulting trace back onto Γ; the converged interface state satisfies $\boldsymbol{g}^\star = T(\boldsymbol{g}^\star)$.

Define the *interface jump* (the fixed-point residual)

$$\boldsymbol{r}^{(n)} \;:=\; T\Big(\boldsymbol{g}^{(n)}\Big) - \boldsymbol{g}^{(n)}. \tag{2}$$

In the Norma implementation (`src/schwarz.jl`) the two terms map to the code variables as

$$\boldsymbol{g}^{(n)} \;=\; \texttt{lambda\_disp} \text{ (from iteration } n{-}1\text{)}, \qquad T\Big(\boldsymbol{g}^{(n)}\Big) \;=\; \texttt{interp\_disp},$$

so that $\boldsymbol{r}^{(n)} = \texttt{interp\_disp} - \texttt{lambda\_disp}$. This is the displacement mismatch the relaxation acts on; it is non-degenerate because `interp_disp` is the freshly projected neighbor trace, whereas the strongly imposed continuity makes the *post-imposition* interface residual identically zero on the Dirichlet side.

The relaxed update common to all variants is

$$\boldsymbol{g}^{(n+1)} \;=\; \boldsymbol{g}^{(n)} + \omega^{(n)}\,\boldsymbol{r}^{(n)} \;=\; \omega^{(n)}\,T\Big(\boldsymbol{g}^{(n)}\Big) + \big(1-\omega^{(n)}\big)\,\boldsymbol{g}^{(n)}, \tag{3}$$

with relaxation factor $\omega^{(n)}$. Classical relaxation uses a fixed $\omega^{(n)} \equiv \omega$; the Aitken variants choose $\omega^{(n)}$ dynamically.

2 Version 1: recursive Irons–Tuck Aitken (`relaxation: aitken`)

With $\boldsymbol{\delta}^{(n)} := \boldsymbol{r}^{(n)} - \boldsymbol{r}^{(n-1)}$, the Irons–Tuck recursion [3] sets

$$\omega^{(n)} \;=\; -\,\omega^{(n-1)}\,\frac{\boldsymbol{r}^{(n-1)}\cdot\boldsymbol{\delta}^{(n)}}{\left\|\boldsymbol{\delta}^{(n)}\right\|^2}. \tag{4}$$

The factor carries the damping history through the multiplicative $\omega^{(n-1)}$, which requires the iterate to have advanced by exactly $\omega^{(n-1)}\boldsymbol{r}^{(n-1)}$ for the recursion to remain consistent (see Section 4).

3 Version 2: non-recursive secant Aitken (`relaxation: aitken secant`)

This is the form derived in the original references, Sambataro–Tezaur [1] (Eq. 9) and Deparis–Discacciati–Quarteroni [2]. With $\boldsymbol{d}^{(n)} := \boldsymbol{g}^{(n)} - \boldsymbol{g}^{(n-1)}$ (the change in the interface iterate) and $\boldsymbol{\delta}^{(n)}$ as above,

$$\omega^{(n)} \;=\; -\,\frac{\boldsymbol{d}^{(n)}\cdot\boldsymbol{\delta}^{(n)}}{\left\|\boldsymbol{\delta}^{(n)}\right\|^2} \;=\; \arg\min_{\omega}\left\|\boldsymbol{d}^{(n)} + \omega\,\boldsymbol{\delta}^{(n)}\right\|^2. \tag{5}$$

The numerator uses the change of the iterate $\boldsymbol{d}^{(n)}$, not the previous residual, and there is no recursive $\omega^{(n-1)}$ factor. Norma forms $\boldsymbol{d}^{(n)} = \boldsymbol{g}^{(n)} - \boldsymbol{g}^{(n-1)}$ directly from the two stored interface iterates rather than from the recursion.

4 Relationship between the two forms

The update (3) gives $\boldsymbol{g}^{(n)} = \boldsymbol{g}^{(n-1)} + \omega^{(n-1)}\boldsymbol{r}^{(n-1)}$, hence

$$\boldsymbol{d}^{(n)} \;=\; \boldsymbol{g}^{(n)} - \boldsymbol{g}^{(n-1)} \;=\; \omega^{(n-1)}\,\boldsymbol{r}^{(n-1)}. \tag{6}$$

Substituting (6) into the secant factor (5) reproduces the Irons–Tuck factor (4) exactly. The two forms therefore coincide *whenever the interface iterate advances by exactly* $\omega^{(n-1)}\boldsymbol{r}^{(n-1)}$. They diverge only where that identity is broken by the initialization and warm-up bookkeeping (Section 6): the secant form reads the actual stored iterates $\boldsymbol{g}^{(n)}, \boldsymbol{g}^{(n-1)}$ and is robust to it, while the recursion silently inherits any inconsistency in the stored $\omega^{(n-1)}$ and $\boldsymbol{r}^{(n-1)}$.

5 An equivalent map-output form of the secant factor

An alternative statement of the acceleration, proposed independently by D. Koliesnikova, works with the differences of the *map outputs* $T(\boldsymbol{g}^{(n)})$ rather than of the iterates. Define

$$\beta^{(n)} \;=\; \frac{\left(T(\boldsymbol{g}^{(n)}) - T(\boldsymbol{g}^{(n-1)})\right)\cdot\boldsymbol{\delta}^{(n)}}{\left\|\boldsymbol{\delta}^{(n)}\right\|^2}, \qquad \omega^{(n)} \;=\; 1 - \beta^{(n)}, \tag{7}$$

followed by the same relaxed update (3), equivalently written $\boldsymbol{g}^{(n+1)} = T(\boldsymbol{g}^{(n)}) - \beta^{(n)}\boldsymbol{r}^{(n)}$. Since $T(\boldsymbol{g}) = \boldsymbol{g} + \boldsymbol{r}$, the map-output difference decomposes as

$$T(\boldsymbol{g}^{(n)}) - T(\boldsymbol{g}^{(n-1)}) \;=\; \boldsymbol{d}^{(n)} + \boldsymbol{\delta}^{(n)}, \tag{8}$$

so that

$$\omega^{(n)} \;=\; 1 - \frac{\big(\boldsymbol{d}^{(n)} + \boldsymbol{\delta}^{(n)}\big)\cdot\boldsymbol{\delta}^{(n)}}{\|\boldsymbol{\delta}^{(n)}\|^2} \;=\; -\,\frac{\boldsymbol{d}^{(n)}\cdot\boldsymbol{\delta}^{(n)}}{\|\boldsymbol{\delta}^{(n)}\|^2},$$

which is exactly the secant factor (5). The map-output form (7) is therefore algebraically identical to Version 2, and in particular is *not* the Irons–Tuck recursion (4): like the secant form, it is non-recursive, reads only stored data, and inherits the robustness discussed in Section 4. That the same factor is reached from two independent derivations is a useful cross-check on both.

Two implementation remarks. First, the storage requirements are the same but the stored quantities differ: (7) retains the two map outputs (or one output and the residual history), whereas the implementation retains the iterate $\boldsymbol{g}^{(n-1)}$ and the residual $\boldsymbol{r}^{(n-1)}$. Second, computing $\omega^{(n)} = 1-\beta^{(n)}$ incurs a cancellation when $\beta^{(n)} \to 1$ (strong damping, $\omega^{(n)}$ small) that amplifies the *relative* error of $\omega^{(n)}$, although the absolute error — which is what enters the update — is unaffected; the direct form (5) avoids even this. The considerations that distinguish the variants in practice are unchanged by the choice of form: the warm-up and safeguard bookkeeping (Sections 6–8) and, most consequentially for dynamics, the choice of which interface variable to relax (Section 8), on which (7) is silent.

6 Initialization

Let $\omega^{(1)}$ be a user-supplied initial factor (Norma: `relaxation parameter`, default 1) and N_0 a warm-up count (Norma: `aitken N0 parameter`, default 1). For $1 \le n < N_0$ the factor is held at $\omega^{(n)} = \omega^{(1)}$; the first dynamic factor is computed at $n = N_0$, once two residuals (Version 1) and additionally two iterates (Version 2) are on record.

7 Numerical safeguards

A single safeguard is retained: a floor on $\|\boldsymbol{\delta}^{(n)}\|^2$ (10^{-20}) that guards the division in (4) and (5) when successive residuals are essentially identical; in that case the previous factor is reused (Version 1) or $\omega^{(1)}$ is used (Version 2). No value clamp is applied to $\omega^{(n)}$: clamping caps the very large or near-zero steps that produce the acceleration and was found to slow convergence, so it is omitted.

8 Which variable to relax: kinematic consistency

For dynamics the interface carries displacement $\boldsymbol{u}$, velocity $\boldsymbol{v}$, and acceleration $\boldsymbol{a}$. Under Newmark integration with a *fixed* step history $(\boldsymbol{u}_n, \boldsymbol{v}_n, \boldsymbol{a}_n)$ and parameters $\beta, \gamma, \Delta t$, the end-of-step kinematics lie on a one-parameter family,

$$\boldsymbol{u}_{n+1} = \boldsymbol{u}_{\text{pre}} + \beta\Delta t^2\,\boldsymbol{a}_{n+1}, \qquad \boldsymbol{v}_{n+1} = \boldsymbol{v}_{\text{pre}} + \gamma\Delta t\,\boldsymbol{a}_{n+1}, \tag{9}$$

with predictors $\boldsymbol{u}_{\text{pre}} = \boldsymbol{u}_n + \Delta t\,\boldsymbol{v}_n + (\frac{1}{2}-\beta)\Delta t^2\,\boldsymbol{a}_n$ and $\boldsymbol{v}_{\text{pre}} = \boldsymbol{v}_n + (1-\gamma)\Delta t\,\boldsymbol{a}_n$. Thus only one of $(\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{a})$ is independent: the integrator's primary solve variable. The relaxation should act on that variable, with the remaining two recovered from (9).

Norma (displacement-form Newmark). The solver unknown is $\boldsymbol{u}$ (`src/time_integrator.jl`, `correct`). The secant variant therefore relaxes *displacement only*, $\boldsymbol{u}^{(n+1)} = \boldsymbol{g}^{(n+1)}$ from (3), and recovers

$$\boldsymbol{a}^{(n+1)} = \frac{\boldsymbol{u}^{(n+1)} - \boldsymbol{u}_{\text{pre}}}{\beta\Delta t^2}, \qquad \boldsymbol{v}^{(n+1)} = \boldsymbol{v}_{\text{pre}} + \gamma\Delta t\,\boldsymbol{a}^{(n+1)}. \tag{10}$$

The legacy `relaxation: aitken` variant instead relaxes $\boldsymbol{u}$, $\boldsymbol{v}$, and $\boldsymbol{a}$ independently with the same $\omega^{(n)}$. A ω-blend of two kinematically consistent triples is again consistent, so the two treatments agree when the incoming triple lies on (9); they differ slightly otherwise because Norma interpolates the three interface histories independently in time. Recovering from a single relaxed variable keeps the imposed interface state exactly on the integrator manifold.

Sierra/SM (acceleration-form). In an acceleration-form integrator the primary unknown is $\boldsymbol{a}$. The same principle then prescribes relaxing the *acceleration* and recovering $\boldsymbol{u}$ and $\boldsymbol{v}$ by inverting (9), i.e. replacing the role of $\boldsymbol{u}$ in (10) by $\boldsymbol{a}$. This is the relevant choice for reproduction in Sierra/SM.

9 Summary for reproduction

Formulation	Relaxation factor	Norma key	Variable relaxed
Irons–Tuck (recursive)	$\omega^{(n)} = -\omega^{(n-1)} \dfrac{\boldsymbol{r}^{(n-1)} \cdot \boldsymbol{\delta}^{(n)}}{\lVert \boldsymbol{\delta}^{(n)} \rVert^2}$	`aitken`	$\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{a}$ (same ω)
Secant (non-recursive)	$\omega^{(n)} = -\dfrac{\boldsymbol{d}^{(n)} \cdot \boldsymbol{\delta}^{(n)}}{\lVert \boldsymbol{\delta}^{(n)} \rVert^2}$	`aitken secant`	primary unknown; recover rest

Table 1: The two Aitken formulations implemented in Norma. Here $\boldsymbol{r}^{(n)} = T(\boldsymbol{g}^{(n)}) - \boldsymbol{g}^{(n)}$, $\boldsymbol{\delta}^{(n)} = \boldsymbol{r}^{(n)} - \boldsymbol{r}^{(n-1)}$, and $\boldsymbol{d}^{(n)} = \boldsymbol{g}^{(n)} - \boldsymbol{g}^{(n-1)}$. For Norma's displacement-form Newmark the primary unknown is $\boldsymbol{u}$; for an acceleration-form integrator (Sierra/SM) it is $\boldsymbol{a}$.

On a representative dynamic non-overlapping Dirichlet–Neumann cantilever problem the secant variant converged in 2 Schwarz iterations per time step, versus 4 to 5 for the recursive variant. The two are not interchangeable there, however: because the displacement-only secant imposes a recovered (rather than independently relaxed) interface velocity and acceleration, its transient differs from the all-fields recursive variant. A full-field comparison against the monolithic single-domain solution shows the recursive variant within 0.7 % while the secant variant is 8.8 % off; the agreement is close only at isolated points such as the cantilever tip. The secant variant thus accelerates the displacement fixed point at the cost of transient accuracy for Dirichlet–Neumann dynamics. For the force-relaxed couplings (Robin–Robin, impedance) no kinematic recovery is involved and all methods reproduce the monolithic solution equally well (0.5 %–0.7 % on the same problem).

References

[1] G. Sambataro and I. Tezaur, *On the role of relaxation and acceleration in the non-overlapping Schwarz alternating method for coupling*, arXiv:2603.17143.

[2] S. Deparis, M. Discacciati, and A. Quarteroni, *A domain decomposition framework for fluid-structure interaction problems*, MOX Report 45, Politecnico di Milano.

[3] B. M. Irons and R. C. Tuck, *A version of the Aitken accelerator for computer iteration*, Int. J. Numer. Methods Eng. **1** (1969) 275–277.